



Register No.

310914822038

JERUSALEM COLLEGE OF ENGINEERING

(An autonomous college affiliated to Anna University, Chennai)

Academic Year 2025-26

Assessment Test-II

Semester IV

B.E. CSE, CSE (AI&ML), CSE (CS), ECE & B.Tech. AI&DS, CS&BS, IT

JMA2402-Applied Linear Algebra

Regulation 2023

TIME : 08.30 am – 10.00 am

DATE: 13.03.2026

MAX. MARKS: 50 marks

PART A (5×2=10)

Q.NO	QUESTION	COURSE OUTCOME	BLOOM'S TAXONOMY
1	If T is a linear transformation of $\mathbb{R}^2$ defined by $T(a, b) = (2a + b, a)$, write the matrix of T relative to the standard basis of $\mathbb{R}^2$	CO4	K2
2	Write the test for diagonalizability.	CO4	K1
3	Define inner product space.	CO5	K1
4	Show that $d(x, y) \leq d(x, z) + d(z, y)$ given $x, y, z \in V$, an inner product space defined over the field F .	CO5	K2
5	State dimension theorem.	CO3	K1

PART B (2×16=32)

6(a)	If T is linear on $V = \mathbb{R}^3$ defined by $T(a, b, c) = (-4a + 3b - 6c, 6a - 7b + 12c, 6a - 6b + 11c)$ find the eigen values of T and an ordered basis β for V such that $[T]_\beta$ is a diagonal matrix.	CO4	K4
OR			
6(b)	Let T on $\mathbb{R}^2$ defined as $T(a, b) = (5a + 4b, a + 2b)$ be a linear transformation. Check for diagonalizability of T and find basis β such that $[T]_\beta$ is a diagonal matrix.	CO4	K4
7(a)	(i) Show that $F^n(\mathbb{C})$ is an inner product space with respect to the inner product defined as $\langle x, y \rangle = \sum_{i=1}^n x_i \bar{y}_i$ where $x = (x_1, x_2, \dots, x_n)$ & $y = (y_1, y_2, \dots, y_n)$	CO5	K4
	(ii) State and prove Cauchy-Schwartz inequality.		
OR			
7(b)	Apply Gram-Schmidt process to obtain an orthonormal basis for $V = P_2(\mathbb{R})$ with inner product defined by $\langle f, g \rangle = \int_0^1 f(x)g(x)dx$. Also find Fourier coefficients of $h(x) = 1 + x$ relative to this orthonormal basis.	CO5	K4

PART C (1×8=8)

8(a)	Prove that $T: M_{2 \times 3}(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$ defined by $T \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} = \begin{pmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 \end{pmatrix}$ is a linear transformation. Also find basis for $N(T)$ and $R(T)$. Compute nullity and rank of T and verify dimension theorem. Also check if T is one-one or onto.	CO3	K4
OR			
8(b)	Prove that $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(a, b) = (a + b, a - b, b)$ is a linear transformation. Also find basis for $N(T)$ and $R(T)$. Compute nullity and rank of T and verify dimension theorem. Also check if T is one-one or onto.	CO3	K4